Predictive Analytics

Lecture 5

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Today

- Exponential smoothing (Chapter 8)
  - What is exponential smoothing?
  - Simple Exponential Smoothing (SES)
    - The smoothing parameter
  - Extensions
    - Holt's Method
    - Holt-Winters Method
    - Damped trend
    - ETS models

Forecasting with exponential smoothing

- Today we focus on exponential smoothing methods
- The basic idea:
  "more recent = more important"

Method	Idea
Naive	Future = last observation
Average	Future = long-run mean
Regression	Future depends on predictors
Exponential smoothing	Future depends mostly on **recent observations**

Forecasting with exponential smoothing

- Suppose we want to forecast a time series, e.g. inflation, weather, demand for iphones….
- Should observations from 10 years ago matter as much as observations from last month?
  - Probably not ☺

Exponential smoothing

- The idea with exponential smoothing is to look at weighted average of past observations – where weights decay **exponentially** as the observation gets older
- I.e., the more recent the observation – the more important it is for our forecast. And the older the observation is – the less importance it has for our forecast

Exponential smoothing

Recent data ⟶ high weight

Old data ⟶ small weight

Simple Exponential Smoothing (SES)

- Suitable when we do not have clear trends or seasonaility in our time series
- We are forecasting the **level**

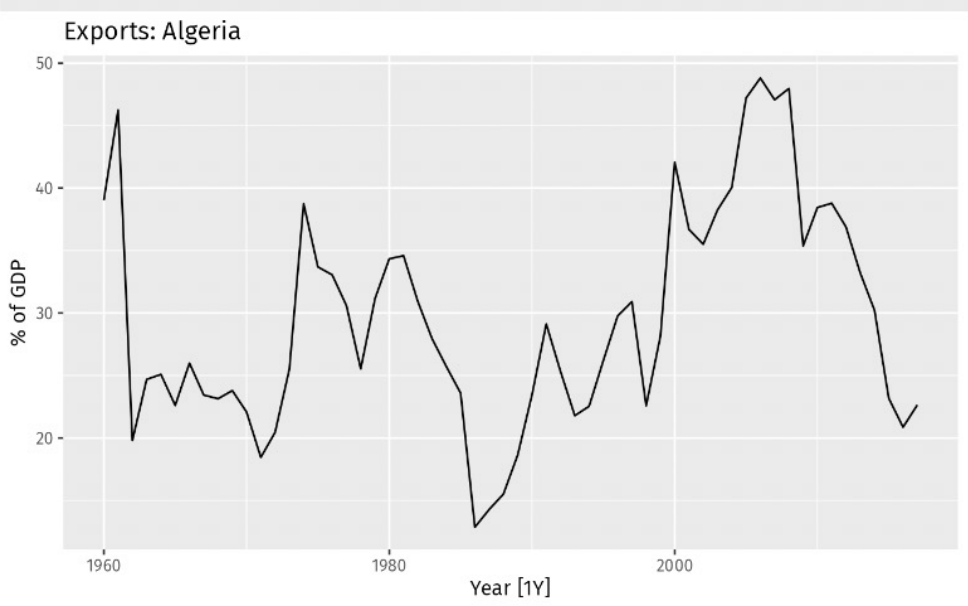


Figure 8.1: Exports of goods and services from Algeria from 1960 to 2017.

Exports and goods from Algeria 1960-2017

What could be an appropriate forecasting model here?

Naive method?

Average method?

SES method?

Simple forecasting

- **Naïve method:** All weight on the most recent observation
- **Average method:** Equal weight on all past observations
- **SES** is between these two extremes:

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$$

- It is a weighted average of lasts period's observation and last period's predicted outcome
- Where $0 \leq \alpha \leq 1$ is the **smoothing parameter** which tells us how much weight we give to the latest data point

The smoothing parameter

- α is the smoothing parameter
- it determines the weight given to the most recent observation)
- If α is high we put much of our emphasis on recent data. This will give us more noise and reactive forecasts
- If α is low we put more emphasis on past values. This will give us a smoother forecast, but one that is slower to react to changes in the data series

Simple exponential smoothing

- SES combines
  - The latest observation
  - The previous forecast
- This means that the forecast evolves gradually over time
- The mode continuously updates the estimated level of the series

SES and past observations

- Note that we can also write SES as a weighted average of all past observations

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1-\alpha)y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \cdots,$$

- Weights decline exponentially
- Older data matters, but less and less

How do we start the process?

- Example: we have data on daily cookie sales
- (Day 1: 100, Day 2: 105, Day 3:103,….)
- We choose the smoothing parameter, e.g. α = 0.3
- We choose a value initial forecast $\hat{y}_1$, usually we just use the first value, i.e. $\hat{y}_1$=100

How do we start the process?

- Now we can find
- $\hat{y}_2 = \alpha y_1 + (1-\alpha)\hat{y}_1$=0.3*100 + 0.7*100 = 100
- $\hat{y}_3 = \alpha y_2 + (1-\alpha)\hat{y}_2$=0.3*105 + 0.7*100 = 101,5
- $\hat{y}_4 = \alpha y_3 + (1-\alpha)\hat{y}_3$=0.3*103 + 0.7*101,5 = 101,95

- etc

	$\alpha = 0.2$	$\alpha = 0.4$	$\alpha = 0.6$	$\alpha = 0.8$
y_T	0.2000	0.4000	0.6000	0.8000
y_{T-1}	0.1600	0.2400	0.2400	0.1600
y_{T-2}	0.1280	0.1440	0.0960	0.0320
y_{T-3}	0.1024	0.0864	0.0384	0.0064
y_{T-4}	0.0819	0.0518	0.0154	0.0013
y_{T-5}	0.0655	0.0311	0.0061	0.0003

The smoothing parameter

Which α do we choose?

- We estimate α using the **training set** by choosing the value that minimizes forecast errors
- Typical procedure

1. We choose some value of α
2. We generate a forecast
3. We compute forecasting errors
4. We try different values of α
5. We select the one that minimizes error (e.g. RMSE)

- But in practice our software estimates the optimal smoothing parameter for us

Is the smoothing parameter likely to be high or low?

- Try to think of two time series

1. One where the optimal α would likely be high
2. One where the optimal α would likely be low

- Can you think of examples? Why is the α likely to be high/low?

Extensions of the SES

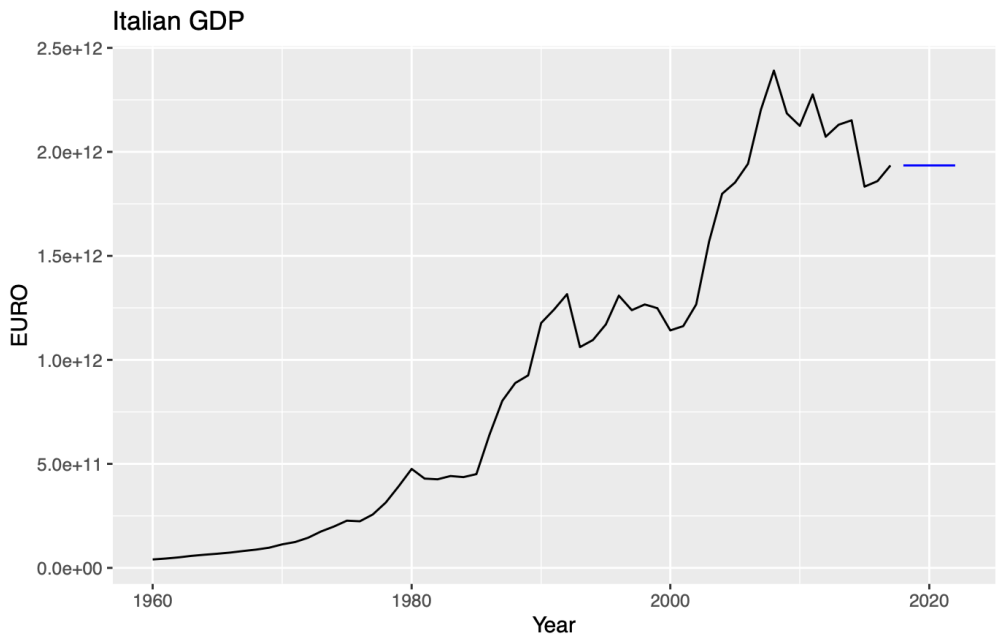
- SES is useful when we have time series that do not exhibit a trend or seasonal patterns
- In cases where we have such dynamics we can use other methods such as **Holt's** or **Holt-Winters**

What if the data has a trend?

In the SES method we assumed we wanted to do a forecast where there was no trend or seasonality

We can see that if our time series has a clear trend, the SES does not do a great job because it only smoothes to the level

Italian GDP



Holt's linear trend method

- Holt's method extends the SES method to allow forecasting of data with a trend
- Now we need to model both the level and the trend:
- $\hat{y}_{t+h} = \ell_t + hb_t$
- Where ℓ_t is the level and b_t is the trend

Holt's linear trend method

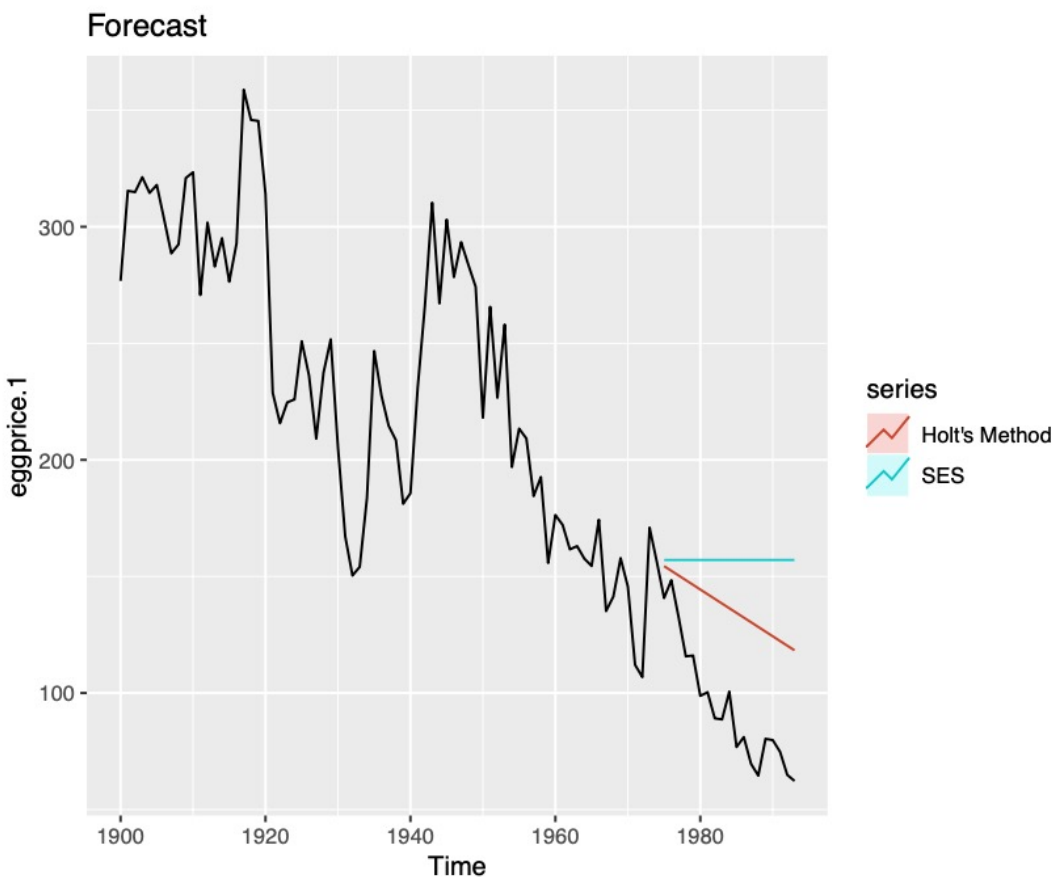
- Level equation: $\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1})$
- Trend equation: $b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}$
- Forecast equation: $\hat{y}_{t+h} = \ell_t + hb_t$

- Level equation: the level is updated using a weighted average of the new observation, and the previous level adjusted by the previous trend
- The trend is also updated based on chanages in the level
- The forecast for *h* periods ahead is given by adding *h* times the trend to the current level (i.e. now the forecast depends on *h*)

Holt's linear trend method

- ℓ_t : is an estimate of the level of the series at time t, and is a weighted average of observation y_t and the forecast for time t
- b_t: denotes an estimate of the trend, or the slope, of the series at time t. It is a weigthed average of the recent change in the series and the previous trend estimate
- α: as before, α lies between 0 and 1, and is the smoothing parameter for the level
- β: is the smoothing parameter for the trend (and also lies between 0 and 1)
  - if β is high, more weight on recent developments, and the model is quick to adjust the trend
  - if β is low, the trend changes more slowly

Example, eggs prices (with training set for 1900-1974)



Holt's method, simple example

- Lets go back to our cookies' example (but change the numbers slightly)
- Day 1: 100, Day 2: 107, Day 3: 105, Day 4: 113,….
- Like before we use our first observations to set the initial values:
- Level, ℓ_1=100. Trend, b_1=107-100 = 7.
- We also need to make assumptions on two smoothing parameters, α and β
- Lets assume α=0.4 and β=0.3

Holt's method, simple example

- Now for day 2, we need to update both the level and the trend

$$\ell_2 = \alpha y_2 + (1-\alpha)(\ell_1 + b_1) = 0.4 \times 107 + 0.6 \times (100 + 7) = 42.8 + 64.2 = 107$$

$$b_2 = \beta(\ell_2 - \ell_1) + (1-\beta)b_1 = 0.3 \times (107 - 100) + 0.7 \times 7 = 2.1 + 4.9 = 7.$$

- The forecast for day 3 is

$$\hat{y}_3 = \ell_2 + b_2 = 107 + 7 = 114$$

Holt's method, simple example

- For the next period we need to update again:

$$\ell_3 = \alpha y_3 + (1-\alpha)(\ell_2 + b_2) = 0.4 \times 105 + 0.6 \times (107 + 7) = 42 + 0.6 \times 114 = 42 + 68.4 = 110.4$$

$$b_3 = \beta(\ell_3 - \ell_2) + (1-\beta)b_2 = 0.3 \times (110.4 - 107) + 0.7 \times 7 = 0.3 \times 3.4 + 4.9 = 1.02 + 4.9 = 5.92$$

- And the forecast in day 4 is then

$$\hat{y}_4 = \ell_3 + b_3 = 110.4 + 5.92 \approx 116.32$$

Damped trend method

- Holt's method assumes that the trend continues for ever, i.e., it gives us a constant trend indefinitely into the future
- This tends to over-forecast, especially for longer forecast horizons
- In many cases there are some bounds on our series, e.g., average time it takes to run a marathon, or the price of eggs. Even if these are decreasing over time, these time series will hit a point where there is little room for a further decrease
- The damped trend method introduces a parameter that **dampens** the trend to a flat line some time in the future

Damped trend method

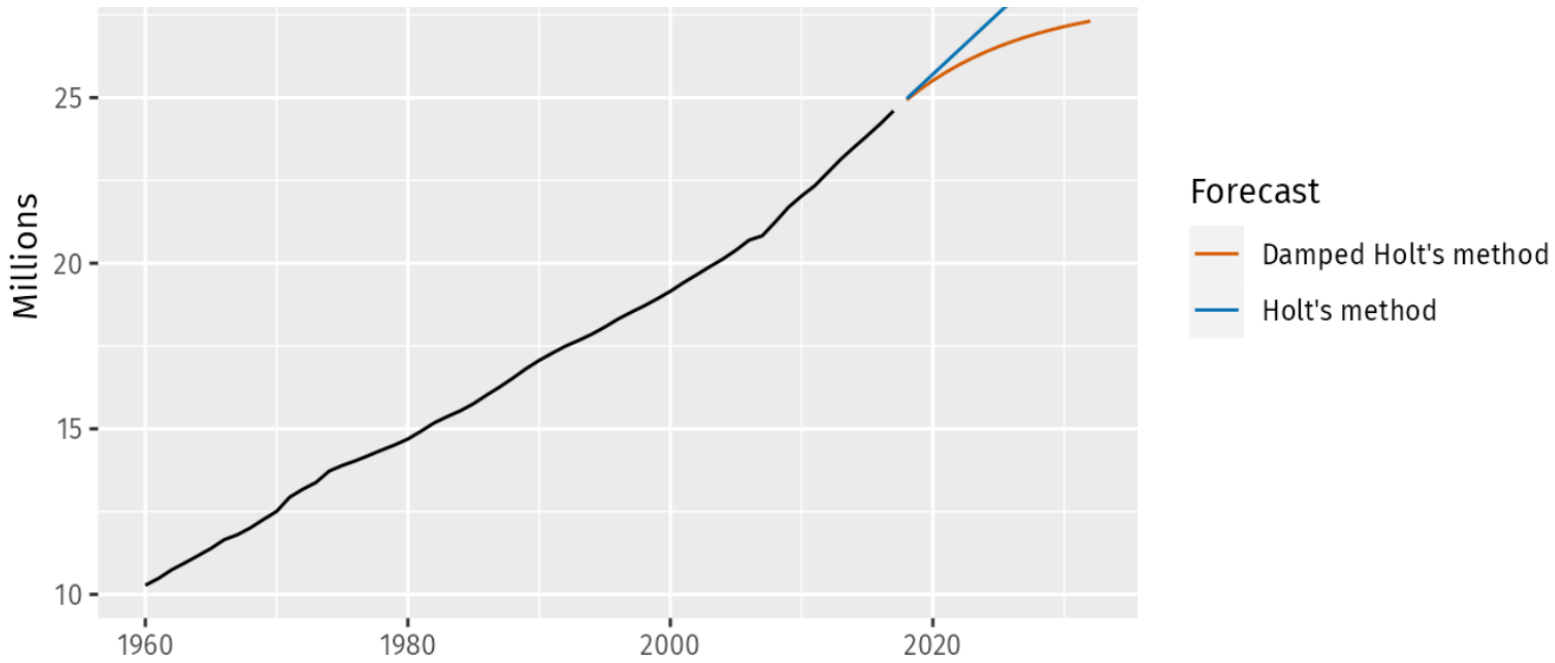
$$\hat{y}_{t+h|}t = \ell_t + (\phi + \phi^2 + \ldots + \phi^h)b_t$$

$$\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\phi b_{t-1}$$

Damped trend method

- The **damping parameter**, ϕ, lies between 0 and 1
- If $\phi = 1$ this method becomes identical to Holt's method
- For values between 0 and 1, ϕ dampens the trend such that it approaches a constant some time in the future
- This means that in the short-run forecasts are trended
- While in the long-run, forecasts are constant



Damped Trend method vs Holt's Method

Methods with seasonality

- The **Holt-Winters' Method** extends Holt's method to capture **seasonality** as well
- Now we add one more smoothing equation – and one more parameter – in the component form representation
- **Two types:**
- Holt-Winters' Additive Method: when seasonal variations are constant
- Holt-Winters' Multiplicative Method: when seasonal variations increase or decrease, proportionally to the level

Holt-Winters' additive method

- **Level Equation:**

  $$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1})$$

- **Trend Equation:**

  $$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}$$

- **Seasonality Equation:**

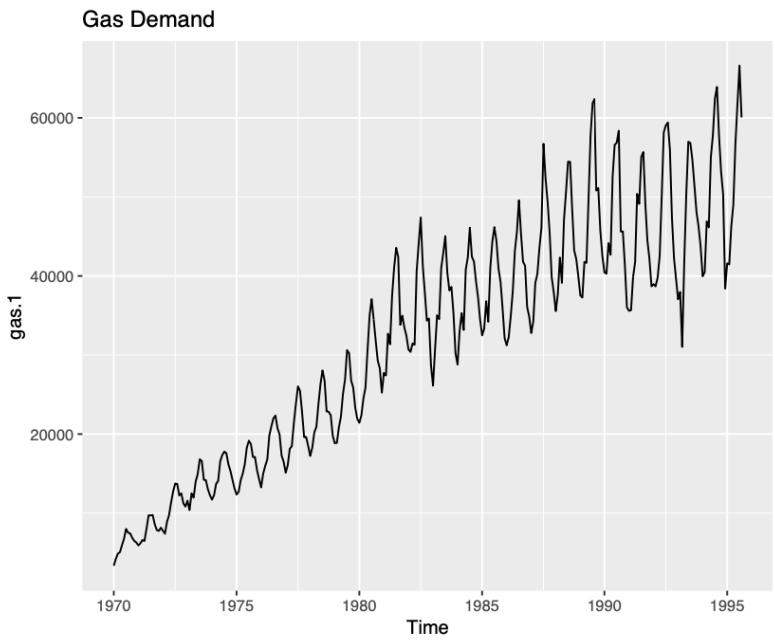
  $$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}$$

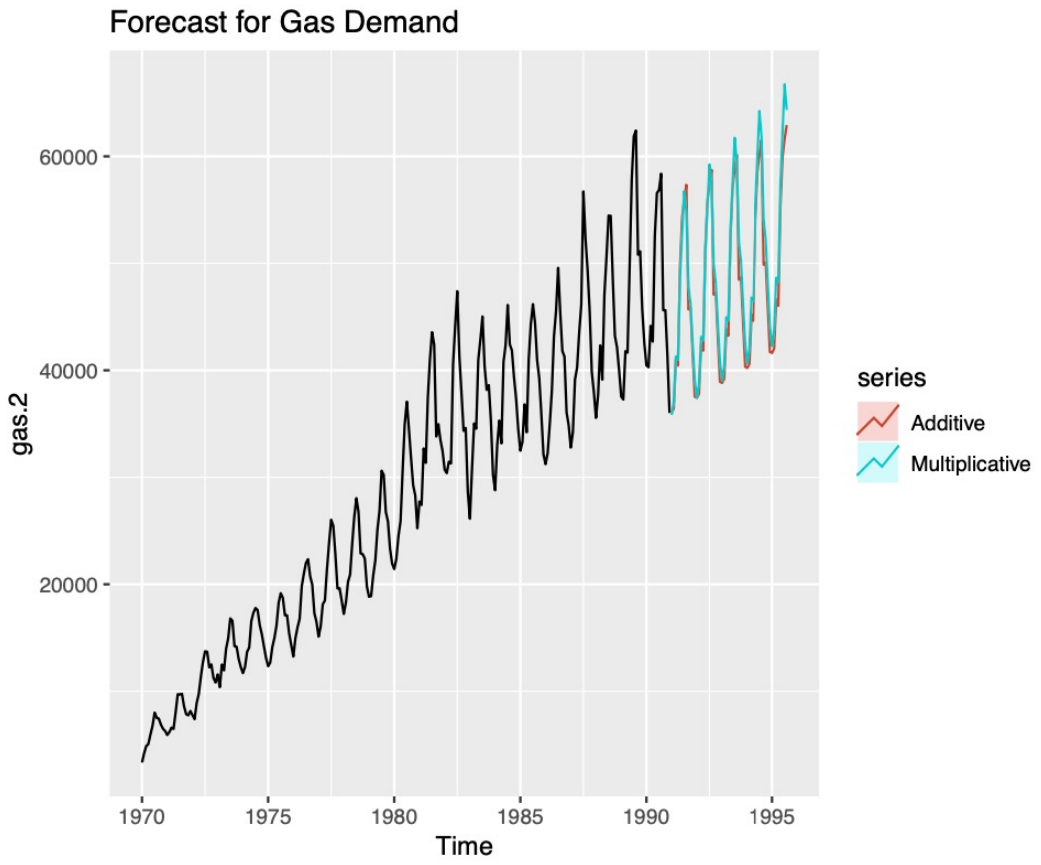
- **Forecast Equation:**

  $$\hat{y}_{t+h} = \ell_t + h\,b_t + s_{t-m+h}$$

Example, demand for gas

Monthly gas data




Forecast for Gas Demand
60000
40000
20000
gas.2
1970
1975
1980
1985
1990
1995
Time
series
Additive
Multiplicative

Exponential smoothing family

Method	Captures
SES	level
Holt (ext. Damped trend method)	Level + trend
Holt-Winters (additive or multiplicative)	Level + trend + seasonality

ETS framework

ETS framework

- ETS stands for Error, Trend, Seasonal
- It is a systematic way to classify exponential smoothing methods
- We can combine different choices for error (additive/multiplicative), trend (none, additive, damped), and seasonality (none, additive, multiplicative)

A taxonomy of exponential smoothing methods

We can apply various combinations of trends and seasonal components

Trend Component	Seasonal Component		
	N	A	M
	(None)	(Additive)	(Multiplicative)
N (None)	(N,N)	(N,A)	(N,M)
A (Additive)	(A,N)	(A,A)	(A,M)
A_d (Additive damped)	(A_d,N)	(A_d,A)	(A_d,M)

A taxonomy of exponential smoothing methods

Note that we have already discussed some of these combinatons

Short hand	Method
(N,N)	Simple exponential smoothing
(A,N)	Holt's linear method
(A_d,N)	Additive damped trend method
(A,A)	Additive Holt-Winters' method
(A,M)	Multiplicative Holt-Winters' method
(A_d,M)	Holt-Winters' damped method

State space models

- For each method there are two models: one with additive errors (A) and one with multiplicative errors (M)
- We label each state space model as $ETS(\cdot,\cdot,\cdot)$ for (Error, Trend, Seasonal)

- Error = $\{A, M\}$
- Trend = $\{N, A, A_d\}$
- Seasonal = $\{N, A, M\}$

- Additive errors: Independent of the level of the series
- Multiplicative errors: Errors scale with the level

As before we want to:

1. Visualize the data
2. Specify the model based on the pattern
3. Estimate the coefficients (using historical data)
4. Evalute the forecast (use training set & test set: compare fitted values to actual observations, using error metrics)
5. Adjust your model if needed, make your forecast

Sales of Ugg boots

If you looked at sales of UGG boots in Denmark since 1995 – how do you think that time series looks like?

If you were asked to forecast the quarterly sales over the next five years, what kind of model/method do you think would be appropriate?

Consider the naïve method, average method and the different exponential smoothing methods

Today

- Key idea: We often want recent observations to weigh more in our forecasting models
- SES models (level)
- Hold model (level + trend)
  - Damped trend
- Holt-Winters models (level + trend + seasonality)
  - Additive vs multiplicative
- ETS provides a unified framework

Next week

- ARIMA models with Thomas Einfeldt
- ARIMA models are forecasting models that focus on autocorrelation, stationarity and differencing

Thank you